It from bit: a concrete attempt

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This work proposes a toy model of the universe based on classical logic, elementary arithmetic, and minimal topological structure, realized as a discrete finite 3-torus augmented by a non-spatial dimension. Each site carries a binary string with ontological significance and evolves according to simple local rules that generate complex behavior, including spherical propagation and harmonic modes. The dynamics exploits both the constraints of toroidal topology and an imposed spherical cavity symmetry. We investigate how such dynamics reproduce features reminiscent of fundamental physical phenomena, providing a novel framework for digital physics. A Platonic seed initializes all state from a simple partition of the non-spatial dimension, after which purely local rules generate constant-speed spherical wavefronts, polarization pairs, and particle-like aggregates. The resulting affinity-driven hierarchy of chiefs, delegates, singletons and propeller pairs gives rise to interaction, inertia and charge quantization without continuous fields or pre-stored physical laws. This approach is not intended as an interpretation of Quantum Mechanics, but as a more primitive descriptive layer.

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"It always bothers me that, according to the laws as we understand them today, it takes a computing machine an infinite number of logical operations to figure out what goes on in no matter how tiny a region of space, and no matter how tiny a region of time. Why should it take an infinite amount of logic to figure out what one tiny piece of space-time is going to do? So I have often made the hypothesis that ultimately physics will not require a mathematical statement, that in the end the machinery will be revealed, and the laws will turn out to be simple, like the checkerboard with all its apparent complexities."

Richard Feynmann

1 INTRODUCTION

John Archibald Wheeler's "it from bit" proposes that the physical world may emerge from discrete information, with bits as the bedrock of reality [1]. Cellular automata (CA)—discrete models of space, time, and local rules—provide a natural setting for exploring this idea, since simple rules can produce complex, physics-like behavior [2, 3, 4].

This work presents a deterministic toy universe built from classical logic, elementary arithmetic, and a minimal 3-torus topology augmented by a non-spatial dimension. Each site carries a fixed-size binary string whose ontological states evolve by local rules, producing spherical wavefronts, harmonic modes, and symmetry-breaking patterns. Wave propagation is confined to a spherical cavity inside the torus, suppressing corner artifacts while preserving periodic boundary conditions. We examine how these emergent structures reproduce features reminiscent of particles, interactions, and spacetime organization. The model is not offered as an interpretation of quantum mechanics, but as a primitive descriptive layer.

Our construction extends foundational work in digital physics: Zuse's discrete universe [3], Fredkin's reversible computation [4], and Wolfram's complex behavior from simple rules [2]. The novelty lies in a layered 3-torus with specific interaction rules that emulate quantum-like and relativistic effects without probabilistic superpositions. The aim is to test whether a deterministic, information-based universe can reproduce key observed phenomena.

The paper is organized as follows: Section 2 begins with a review of previous work in the area, followed by Section 3 with the foundational stage, defining the universal fabric. Section 4 builds the initial state. The light frame is presented in Section 6. Section 8 defines particles, followed by Section 7 that illustrates bubble interactions and explores the role of charge conjugation. Section 10 provides key results. Section 11 succinctly contemplates the bridge between the model and the QM formalism, and Section 12 outlines potential future directions. Finally, Section 13 offers a summary and a synthetic interpretation of the model.

2 RELATED WORK

Stanislaw Ulam and John von Neumann

Stanislaw Ulam's lattice models of crystal growth led him to suggest a discrete framework for self-replication to John von Neumann, who then developed the first cellular automaton—a 29-state system on a 2D grid—laying the groundwork for CA theory [5, 6].

Konrad Zuse

Konrad Zuse was the first to propose a CA-based universe [3]. In *Rechnender Raum* (Calculating Space) he argued that space and time are discrete and that physical processes arise from local, rule-based state transitions on a grid of finite-state machines. He reformulated classical physics in algorithmic terms, introduced "digital particles" as localized, self-reproducing lattice patterns, and explored how higher-dimensional grids and Lorentz invariance might emerge from discrete dynamics. Although a heuristic program, his work established digital physics as a research direction and influenced later work by Fredkin and Wolfram.

Edward Fredkin
Fredkin [4] extended Zuse's vision with the Finite Nature Hypothesis: the universe is a CA whose evolution preserves information. Reversible computation is central to his proposal, and the Fredkin gate is a concrete reversible logic primitive. Partitioned CA and lattice-gas models show that discrete, local, invertible rules can conserve mass, energy, and momentum while reproducing continuum equations at coarse-grained scales, supporting the idea that CA-based frameworks can capture key features of physical law.

Philosophical implications:

- The universe is not merely like a computer—it *is* a computer.
- Particles are patterns of information, not substances.
- Physical laws may be *emergent programs* running on a CA-like substrate.

Fredkin's ideas influenced Margolus, Wolfram, and others.

Martin Gardner
M. Gardner popularized Conway's Game of Life in *Scientific American*, drawing broad interest to CA and complex systems [7].

Norman Margolus
Margolus [8] modeled physical phenomena as discrete, reversible, strictly local computations. To avoid the irreversibility of standard CA updates, he introduced block cellular automata (the Margolus neighborhood), where permutations on shifting blocks conserve mass, energy, and momentum directly from bit-level dynamics. This work bridged computation theory and physics, influencing reversible computing, digital physics, and quantum CA.

Gerard 't Hooft
Gerard 't Hooft [9] proposed a deterministic CA foundation for quantum mechanics. In his "Cellular Automaton Interpretation of Quantum Mechanics," quantum states are coarse-grained equivalence classes of deterministic ontological CA states, and the Hilbert-space formalism can be reconstructed from classical CA dynamics.

Stephen Wolfram
Wolfram's *A New Kind of Science* [2] argues that simple CA rules can generate complex behavior and even be Turing complete (Rule 110). His main claims are:

1. **Simple rules, complex behavior:** Even simple CA rules can produce unpredictable, complex patterns.
2. **Rule 110 and universality:** Rule 110 is Turing complete, suggesting computation is fundamental.
3. **Computational universe hypothesis:** The universe may evolve like a simple program.
4. **Causal networks and hypergraphs:** The Wolfram Physics Project [10] models space as a hypergraph updated by rewrite rules.
5. **Principle of computational equivalence:** Most nontrivial systems are computationally equivalent.

The Wolfram Physics Project generalizes CA from grid cells to hypergraph nodes; it remains incomplete as of 2025.

3 THE UNIVERSAL FABRIC

Let L be a large odd integer, divisible by 3. The fundamental fabric of this toy universe is a lattice of cells arranged in a 3-torus ($\mathbb{T}^3$) topology. This finite, closed, and discrete space is periodic: each boundary wraps around to connect with its opposite edge, ensuring global continuity without physical boundaries.

In addition to the three spatial dimensions, the framework includes a compact non-spatial dimension composed of W discrete layers, each containing exactly L^3 cells.

Rather than representing an additional physical direction or a momentum space, the W dimension is an internal dynamical space that continuously organizes the elementary excitations (referred to as *bubbles* and formally defined in Sect. 6.2) through pairing and reorganization processes. Every bubble possesses both a spatial position in the three-dimensional torus and an internal coordinate in W. While the spatial coordinates evolve through propagation in $\mathbb{T}^3$, the internal dynamics continuously modifies the pairing relationships among bubbles.

The internal organization in W is essential for the continuous dissociation and reconstruction of collective structures. Stable three-dimensional particles emerge as coherent islands whose constituent bubbles remain correlated in physical space. During scattering or annihilation events these islands may completely dissolve into individual bubbles, which subsequently reorganize through new pairing processes, allowing new collective structures to emerge while conserving the total bubble population.

The value

$$W = 3L^2$$

defines the size of the internal dynamical dimension. Its quadratic scaling naturally matches the characteristic cross-sectional scaling of the discrete three-torus and provides sufficient internal degrees of freedom for the continual pairing, dissociation and reorganization of bubbles throughout the evolution of the cellular automaton. A dynamical motivation for this scaling, together with the resulting total bubble population $3L^3$ and the $9L$ W-islands, is given in Section 9.

3.1 The cell structure

Each cell carries the same fixed-size bit string, formatted as shown in Table 1.

Evolution progresses in uniform steps. The input and output ports of each cell are conceptually linked through a Finite State Machine. This design uses integers and simple logical rules, avoiding complex mathematical constructs.

Physical properties

The more intuitive variables are grouped here.

- **Charge (ch)**
  Defined by bit constants q, w_1, w_0, c_2, c_1, c_0. Charge encompasses various discrete properties associated with the particles, derived from specific bit values that define their interactions. They are the main responsible for the emergence of forces.

- **pBit (pB)**
  This bit marks the local in-phase sign of the emergent polarization pair $(\mathrm{pol}_u, \mathrm{pol}_v)$ reconstructed from the broadcasted arrival stamp $b(x)$: it is true when the in-phase component pol_u is positive, i.e. $pB = (\mathrm{pol}_u > 0)$. It triggers the electric interaction channel.

- **sBit (sB)**
  This bit signals the emergent transverse polarization of the cell. It is true when the reconstructed transverse component pol_v is positive, i.e. $sB = (\mathrm{pol}_v > 0)$; the pair is derived from the broadcasted arrival time of the elected momentum vector (Sect. 5.3). No spiral path is precomputed.

- **Affinity (a)**
  Groups layers into particles, facilitating the aggregation of smaller units into more complex entities. The affinity integer variable is essential for the formation of composite structures; Without it, only a holistic universe would emerge. It is also the substrate for the emergent phenomenon of entanglement and characterizes the track left by the particles in self-interference.

Wavefront shaping

This set of variables and constants forms the mechanism for nearly perfect spherical wave propagation (with discrete constraints).

- **Dynamic squared radius (r^2)**
  The squared Euclidean distance from the current wave source, updated locally by an incremental rule through the 6-neighborhood. It is INF before the wavefront reaches the cell.

- **Dynamic radius** (r)
  r is the integer radius propagated alongside r^2 and corrected by local square-boundary tests; it is used in the wave equation, shell forcing, and boundary absorption.

- **Wavefront active** ($active$)
  This bit is true when the cell currently lies on the pulsating spherical wavefront. It is computed by comparing the cell's propagated integer radius r with the local light-frame radius $\mathrm{effective\_t}(t)$, which advances one cell per light frame; it is not precomputed and requires no square root.

- **Wave phase** (f)
  This integer stores the local phase of the wave oscillator. In the current implementation the sieve bit $s2B$ is computed directly from the wave displacement u and the step counter t.

- **Wavefront tick** (t)
  This long integer provides a clocking mechanism for activities that depend on the spread of information, wavefront or interactions. It is ultimately a light step counter.

In addition, the wave field uses the wave displacement u and velocity v. The sieve bit $s2B$ is computed at each wave step from u and the step counter t by a Bresenham-like test and is then gated by the active flag.

The wave phase f and the wavefront tick t advance together; the test produces sieve triggers at a rate proportional to the wave amplitude.

Operational variables
The model performs operations—translation, convolution, and wavefront propagation—that must be coordinated across the lattice and therefore may appear nonlocal. Such coordination is supported by the distinction between two clocks. The housekeeping tick k is incremented synchronously in every cell (all cells share the same value) and therefore acts as an atemporal, global phase clock for the FSM. The wavefront tick t, on the other hand, is a local physical clock: it is reset at each source and may differ across layers and regions, carrying the actual light-step evolution. The interplay between the common k and the local t is what allows the framework to schedule seemingly nonlocal processes without introducing any superluminal signal.

The variables defined below are the essential components of this two-clock scheme, managing the dynamics of translation (this massive shift of data inside a layer is considered in Sect. 6.6), convolution (see Sect. 6.4), and wavefront propagation.

- **translation offset** (**c**)
  This integer vector determines the reissue address, guiding the position where an element is reallocated within the system. This variable controls the adjustment needed to realign or reposition bubbles in the lattice. The displacements follow the general rule

  $$d_i = (t_i - c_i + L) \bmod L,$$

  where:

  $$\begin{aligned} d_i &: \text{positive displacement along axis } i \ (i \in \{x, y, z\}), \\ t_i &: \text{coordinate of the target point along axis } i, \\ c_i &: \text{coordinate of the central point along axis } i, \\ L &: \text{length of the grid's side (odd)}, \\ \mathrm{mod} &: \text{modulo operation.} \end{aligned}$$

  For an odd L, the coordinate of the central point is:

  $$c_i = \frac{L-1}{2}.$$

- **Housekeeping tick** (k)
  This long integer* is an atemporal clocking mechanism, used for maintaining internal processes.

* In this context, "long" refers to a scale on the order of L^2.

- **Sieve test result** ($s2B$)
A Boolean bit seeded at each wave step from the local wave displacement u and the step counter t, and propagated during interactions. For $u > 0$ on an active cell, the deterministic test

$$\big((u \cdot t) \bmod \text{SHELL\_TARGET}\big) < u$$

produces a trigger, where SHELL_TARGET $= 16384$ is the reference amplitude scale; the stored value is gated by the active flag, $s2B = \textit{active} \wedge \textit{trigger}$. During electroweak interactions it is propagated as $s2B' = s2B \wedge \textit{active}$. Electroweak interaction only happens when this bit is true.

- **Collapse flag** (kB)
A variable bit indicating the occurrence of collapse (defined in Sect. 6) in the interaction. It is used to warn the components of a particle that a collapse occurred. Electroweak collapse only happens when this bit is true.

- **Blob flag** (bB)
A variable bit indicating whether the pair can participate in a blob (a group of superposed pairs with the same affinity, same bB bits and the same active pB).

We highlight the difference between k and t. While k represents the fundamental timing, being updated simultaneously in all cells at all times—each having the same value and wrapping modulo—variable t is updated at each k period and can take different values across layers and within different regions of a layer. Variable t better approaches the notion of physical time.

3.2 The host lattices

The space described above is occupied by three coexisting *3D*-Euclidean spatial lattices made of L^3 cells each, repeated in W layers (that is, $3 \cdot 3 \cdot L^5$ cells total). All cells within each layer pulsate in unison, incrementing alternately the couple of evolution counters (k and t) and exchanging information with their seven neighbors (six in dimensions x, y, and z , and one in dimension W, which facilitates the transfer of information between lattices and detects interactions). This exchange occurs alternately, ensuring time homogeneity is recovered every two clock ticks, grouped in four passes (see further details below), thus configuring a cellular automaton.

3.3 Purity constraints

The rules of this toy universe are deliberately constrained so that the model remains a *pure* cellular automaton. These constraints are not implementation details; they shape the physics that the automaton can produce and are recorded here as design criteria for all sections that follow.

Locality. A cell's next state is computed only from its own current state and from the states of a fixed, finite neighborhood of adjacent cells. No cell reads global counters, hidden pointers, queues, frontier lists, or pre-computed tables that depend on the whole lattice.

Synchronicity and parallelizability. Every cell updates simultaneously from the *current* lattice into a second, identical lattice (`grid` and `grid_next`). The two buffers are the only shared memory; after the tick the roles are swapped. This makes the rule directly parallelizable: any subset of cells can be updated independently given the same neighbor window.

Integer-only runtime. Inside the transition rule all calculations use only addition, subtraction, bit shifts, and comparisons. Multiplication, division, floating-point arithmetic, square roots, trigonometric functions, and any other continuous operation are forbidden in the runtime loop. Constants that depend only on the fixed axis or on topological bounds may be computed once at initialization, but they remain read-only thereafter and must themselves be integers.

No tables. No pre-computed lookup tables encode the dynamics. Distances, phases, and interaction weights must propagate through local rules from the initial seed, not be read from a global array. The only permissible constants are the small integer parameters of the automaton (e.g. `CONVOL`, `DIFFUSION`, `RELOC`, `REISSUE`, `FLOOD`) and the read-only axis constants; these are topological data, not tabulated state.

Name	Type	Symbol	Obs.
			Physical properties
charge	B6	ch	bits q, w_1, w_0, c_2, c_1, c_0
pBit	B	pB	local in-phase sign, $pB = (\mathrm{pol}_u > 0)$
sBit	B	sB	emergent transverse polarization bit, $sB = (\mathrm{pol}_v > 0)$
affinity	N	a	groups bubbles into particles
position	V	$\boldsymbol{x}$	spatial coordinates
			Wavefront
dynamic squared radius	DN	r^2	CaRaSh-style squared distance from the wave source
dynamic radius	DN	r	integer Euclidean radius from the source
wavefront active	B	$active$	true when $r = \mathrm{effective\_t}(t)$
wavefront tick	DN	t	light clocking
wave phase	DN	f	local wave phase; the sieve bit $s2B$ is computed from u and t
broadcast stamp	DN	$b(x)$	arrival tick of the elected-momentum news (0 = never reached)
polarization u	N	pol_u	in-phase reconstructed polarization component
polarization v	N	pol_v	transverse reconstructed polarization component
			Operational variables
translation offset	V	$\boldsymbol{c}$	determines reissue address
momentum vector	V	$\boldsymbol{m}$	stable reissue/translation direction (unit Cartesian sign; the helical broadcast uses
translation impulse	V	$\boldsymbol{reloc}$	signed displacement applied to the source center after interaction; consumed in o
housekeeping tick	DN	k	timeless clocking
sieve test result	B	$s2B$	seeded from u and t, propagated with the active flag
sB hunting	B	hB	flag to find the sB bit
collapse flag	B	kB	hints collapse
blob flag	B	bB	marks pairs eligible for blob formation

TABLE 1
Cell structure.

The formatted bit string is valid for all cells, where *N* represents a integer, *DN* a double integer, *V* a three-dimensional integer vector, and *B* a boolean. The conventions for the values *false* and *true* for each charge bit are as follows: q: **+**, **-**; w_1: [O]rbis, [D]ark; w_0: [R]ight, [L]eft; c_2: [B]lue, anti[$\bar{B}$]lue; c_1: [G]reen, anti[$\bar{G}$]reen; c_0: [R]ed, anti[$\bar{R}$]ed (see Sect. 3.1).

Finite-state and deterministic. Each cell contains a fixed-size bit string. The transition rule is a finite function of this string and its neighborhood, with no Turing-complete constructs, no unbounded iteration, and no randomness.

Occam's razor. The cell state is kept minimal. Fields such as `active`, `spin`, or the wave counters are not decorative metadata; each must be derivable from, or necessary to, the local rule. Auxiliary global variables, linked lists, buckets, or index arrays that point into the lattice are rejected because they would make the automaton's state depend on the order of evaluation rather than on the cell contents alone.

Simple initialization. The initial state is set by fiat only at $t = 0$: a small seed is planted in an otherwise blank lattice. All subsequent structure—wavefronts, polarization pairs, charges, and particle-like aggregates—emerges from the local rule, not from hand-tuned initial data.

The present model is finite and defined on a 3-torus of side L, so L and $W = 3L^2$ enter as *topological data* (the size of space and of the internal dimension), not as free parameters of the rule. Because the rule itself uses only the pre-computed integer bounds above, the dependence on L remains at the level of topology; the transition function is still local, homogeneous, and free of size-dependent case distinctions. Consequently, the model is not scale-invariant, but it is an L-parameterized finite CA in the sense surveyed by Das and Sikdar [11]: the lattice size fixes the arena, not the logical form of the transition.

The distance between cells along the three spatial axes corresponds to the fundamental quantity X in the real world (calculated in Appendix A).

The three lattices mentioned above are referred to as *main*, *draft*, and *mirror*. The main lattice is the current lattice used in interactions; draft is the modified data; and mirror is used in comparisons of information in different W addresses, it is just a snapshot for comparison - it doesn't evolve during the diffusion/translation phases. A cell is capable of modifying its own draft cell, but not itself.

The cell bit string, together with the inter-cell rules/gates, forms the basic ontological object in this model.

3.4 Charges

Charges, or more precisely, fundamental charge fragments, are constant-valued bits (*ch* in the above table) that govern the interactions among particles. Their properties draw inspiration from the Standard Model of particle physics. All cells in a layer have the same charge value.

The *electric* charge q is associated as ever with attraction/repulsion, while the *weak* charge with its two bits w_1 and w_0, or *chirality*, is associated with congruence. Since the universe is closed, the addition of an extra bit guarantees this sense of congruence. The sector with $w_1 = 0$, will be called *Orbis,* while the sector with $w_1 = 1$ will be called *Umbra*. On the other hand, the three *color* charges c_2, c_1, c_0 enforce the strong force structure. Color codes are $N : 000$, $R : 001$, $G : 010$, $\bar{B} : 011$, $B : 100$, $\bar{G} : 101$, $\bar{R} : 110$, $\bar{N} : 111$ for *neutral*, *red*, *green*, *antiblue*, *blue*, *antigreen*, *antired* and *anti-neutral*, respectively. Let signature be $sig = c_2 + c_1 + c_0$. A bubble is matter M if $sig < 2$, or antimatter $\overline{M}$, otherwise. Also, it is neutral N, if $sig = 0$ or anti-neutral $\overline{N}$, if $sig = 3$.

The pBit pB marks the positive in-phase half-cycle of the polarization pair, while the sBit sB records the transverse sign derived from the reconstructed $(\mathrm{pol}_u, \mathrm{pol}_v)$ pair. They are not precomputed path patterns; both emerge from the cellular dynamics.

Given the basic structure of our model, we shall now proceed to populate it with initial information.

4 INITIAL STATE

The initial state is the only information set by fiat. It fixes the static, non-dynamical configuration of every cell at $t = 0$ and is copied to the draft lattice as well. All dynamic quantities—the wavefront radius, the wave cloud, the polarization pair $(\mathrm{pol}_u, \mathrm{pol}_v)$, and the interactions—are generated from this seed by the same local transition rule.

4.1 Internal addressing and W-islands

The internal non-spatial dimension has size

$$W = 3L^2, \tag{1}$$

where L is a large odd integer divisible by 3. During the Platonic Initialization, W is partitioned axiomatically into

$$N_I = 9L \tag{2}$$

identical W-*islands*, each containing

$$\ell = \frac{W}{N_I} = \frac{L}{3} \tag{3}$$

positions. A W-island is an internal charge family: its ℓ copies share the same six charge bits but have distinct pB/sB patterns. The distinct polarizations provide the internal angular diversity needed for the Hofer effect (Sect. 12.9). The detailed dynamics of how these copies aggregate into $1K + nD$ islands, and how the island population stabilizes, is treated in Section 9.

4.2 The Platonic seed

Among the many possible choices, we select the following deterministic Platonic seed:

- The spatial address $\boldsymbol{x}$ is set to the appropriate Cartesian positive coordinates $(0 \ldots L-1)$. Each source center is placed on one of the six Cartesian face positions around the lattice center, e.g. $(-L/2, 0, 0)$ relative to the center, so that the bubbles are spread isotropically without random numbers.
- The internal address w is immutable; the charge bits w_0, w_1, electric charge $q = w_1 \oplus w_0$, and color are derived from w as described above.
- The affinity a and the auxiliary leader identity $leader\_w$ are initialized to the island index of the W-island to which w belongs (Sect. 4.1). A true chief (K) emerges later through convolution, as described in Section 9.
- The housekeeping counter k and the wavefront counter t are set to zero; the contraction/phase vectors $\mathbf{c}$ are zero and $s2B$ is reset.
- The momentum-direction vector $\boldsymbol{m}$ is assigned a fixed unit step along one Cartesian axis. Consecutive layers share the same axis and receive opposite signs according to the electric charge bit $q_i = w_1^i \oplus w_0^i$, so that $q = 1$ points in the positive direction and $q = 0$ in the negative direction. With $W = 3L^2$ and L odd, this leaves the most perfect antipodal balance possible.
- The consumable translation impulse $reloc$ is zero.
- The wavefront data are seeded only at the source centers: $u = v = 0$ and $r^2 = 0$, with a fixed amplitude U_0; all other cells have $r^2 = \text{INF}$ and $u = v = 0$.

Real numbers and high-level functions are used only as intermediaries to compute these integer parameters; they are not stored in the lattice. The dynamics is therefore strictly integer-based.

4.3 What the seed does not contain

The seed contains no precomputed radius, no wave cloud, no $(\text{pol}_u, \text{pol}_v)$ pair, no $pB/sB/s2B$ pattern, no interactions, and no randomized data. Those are the first emergent consequences of the seed and are described in Sect. 5.

4.4 The initial thermal bath

Before the dynamic patterns—the vectors $\boldsymbol{m}$, the charge islands, and the wave cloud—stabilize, the evolution is chaotic. This early mixing is not a defect: it acts as a deterministic "thermal bath" that contributes to the apparent randomness of particle physics once the stable attractors are reached.

5 PHASE STEP AND EMERGENT WAVE FIELDS

Once the Platonic seed is fixed, the first CA ticks generate the wavefront geometry that feeds the light-frame FSM. The subsections below describe those emergent fields.

5.1 Dynamic radius and the Euclidean metric

The wave field is seeded at each of the six source centers placed on the Cartesian faces around the lattice center. From each source, the wavefront expands as a sphere in the cubic lattice, so we need a local, integer-valued approximation of the Euclidean distance from every cell to its source.

Squared-distance update. Let the source have integer coordinates (c_x, c_y, c_z) and let (x, y, z) be the cell coordinates, all taken modulo L along each axis of the 3-torus. The exact squared Euclidean distance is

$$r^2 = (x - c_x)^2 + (y - c_y)^2 + (z - c_z)^2.$$

In the 6-neighborhood (axis-aligned moves only), an outward step along the axis with absolute offset a from the source changes r^2 by

$$\Delta r^2 = (a + 1)^2 - a^2 = 2a + 1.$$

Therefore the local update is purely additive:

$$r^2_{\text{new}} = r^2_{\text{old}} + 2a + 1,$$

where a is the cell's own absolute coordinate along the moved axis. Each cell needs only its stored r^2 and its absolute offset from the source, both integers. The update is performed synchronously over the whole lattice at every tick: each cell writes its new value into a second, identical lattice (`grid` and `grid_next`), so the rule is a local double-buffer CA transition. In one tick a cell can only learn from its six immediate neighbours, so the reachable region grows by one lattice layer per tick, exactly as a light-cone should; before the wavefront reaches a cell, $r^2 = \text{INF}$. Because the edge weights are non-negative, repeated synchronous sweeps converge to the same minima a Dijkstra/BFS relaxation would produce, but without any global queue or frontier list.

Radius correction method Several physical formulas require the ordinary radius r rather than its square: the wave phase $(r \cdot 2R) \bmod \text{phase\_full}$, the wave envelope $\sin(r)/r$, and the radial shell count used for the $r \cdot \sin(r)$ density all use r directly. The lattice is integer-valued, so r is the largest integer not exceeding the true Euclidean distance, $r = \lfloor \sqrt{r^2} \rfloor$. We propagate r alongside r^2. When a cell with radius r passes the wavefront to a 6-neighbor, the neighbor's squared distance increases by $2a+1$, which changes the true radius by 0 or 1. Hence the neighbor's integer radius is either r or $r + 1$, and the correct choice is the one that satisfies

$$(r')^2 \leq r^2_{\text{new}} < ((r') + 1)^2.$$

This test uses only integer additions and comparisons. A one-step source-center motion changes any cell's radius by at most one, so a tiny correction loop around the previous r is enough to keep the value exact after the source moves.

The discretization error is at most one lattice unit and does not accumulate, because r^2 is re-anchored by the source center each sweep and the radius is re-corrected from the exact r^2.

Efficiency of the method Extracting r once per cell per tick would require a bounded loop of bit operations. By propagating r together with r^2 we replace that generic root extraction by a single local square-boundary test: the parent cell already knows its own radius, and the child radius is either the same or one larger. The source center re-anchors $r = 0$ each sweep, so errors cannot drift. This keeps the FSM transition small and avoids carrying a full integer square-root routine in the inner loop.

Why the 6-neighborhood? The 6-neighborhood corresponds to the three Cartesian axes and is the minimal set of edges compatible with the discrete rotation/reflection symmetries of the cubic lattice. Edge weights $(2a + 1)$ then give the correct squared Euclidean increments for those edges. Moves along face diagonals or body diagonals would require larger neighborhoods and non-integer corrections; the 6-neighborhood together with the squared-distance update already captures the Euclidean metric without adding extra edges.

Boundary behavior. Because the lattice is a 3-torus, the distance field wraps through the periodic boundaries. The update remains local: a step that crosses a face of the cube uses the periodic coordinate and the same $2a + 1$ increment. As long as the source center is taken modulo L in each axis, the resulting r^2 is the squared distance on the torus. Multiple source centers are handled independently, one per W-layer, and the local relaxation within each layer does not mix their fields.

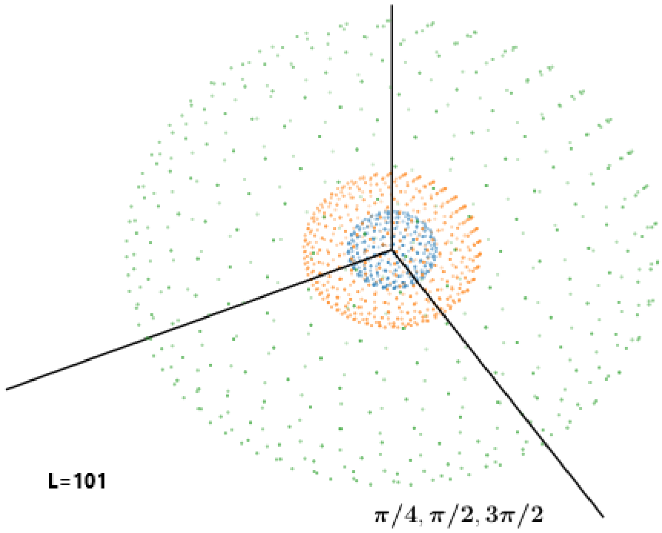

FIGURE 1
The emergent $r \cdot \sin(r)$ wave cloud.

The cell-by-cell wave amplitude follows $\sin(r)/r$, while the number of cells per spherical shell scales as r^2; their product gives the radial interaction density $r \cdot \sin(r)$ used for electromagnetic coupling.

5.2 The wave cloud

The wave cloud is not a precomputed mask. It emerges from a discrete wave field that propagates outward from each bubble. Each cell carries a wave amplitude that evolves according to an integer-only wave equation, while a pulsating spherical wavefront marks the cells whose propagated integer radius r equals the local light-frame radius $\text{effective\_t}(t)$ as *active*. The sieve bit $s2B$ is computed directly from the local displacement u: for $u > 0$,

$$\big((u \cdot t) \bmod \text{SHELL\_TARGET}\big) < u, \qquad \text{SHELL\_TARGET} = 16384,$$

sets a trigger, and the result stored in the cell is $s2B = \mathit{active} \wedge \mathit{trigger}$. During interactions the bit is propagated as $s2B' = s2B \wedge \mathit{active}$. The coincidence of *active* cells whose $s2B$ bit is true forms the sieve.

Because the number of cells at radius r on a cubic lattice scales approximately as r^2, the radial density of active cells with $s2B$ true follows $r \cdot \sin(r)$, which is the electromagnetic coupling density (see Figure 1). The underlying wave amplitude per cell is $\sin(r)/r$, so the $r \cdot \sin(r)$ density is a geometric consequence of binning active $s2B$ cells over spherical shells. The $s2B$ bit propagates this test between bubbles.

The cloud is generated on the fly inside each light frame. At $t = 0$ the wave field is zero except for the excitations of fixed amplitude U_0 with $r^2 = 0$ at the six source centers; all other cells have $r^2 = \text{INF}$ and $u = 0$. The constant U_0 is chosen independently of L (for example, $U_0 = 2048$ in the reference implementation), so the same seed remains valid as the lattice grows; in the $L \to \infty$ limit it becomes a point excitation on an unbounded lattice. As the light-frame radius expands and contracts one cell per frame, the normalized wave amplitude yields the $\sin(r)/r$ envelope per cell, and the active-triggered cells form the $r \cdot \sin(r)$ density. The wave phase f, the Bresenham accumulator, and the active flag are therefore dynamical, not imprinted by fiat.

5.3 Emergent polarization pair $(\mathrm{pol}_u, \mathrm{pol}_v)$

Each active cell carries an integer pair (u, v), stored as pol_u and pol_v, that represents the transverse polarization in the plane tangent to the spherical wavefront. The pair is not a geometric function of the local radius; it is derived from the *broadcasted arrival stamp* $b(x)$ *of an elected momentum vector*. At every turn-around of the breathing wavefront the automaton elects one direction out of its own state, a helical walker broadcasts the news of that election across the layer, and every cell converts its private arrival stamp into a polarization phase. The mechanism has three stages—election, broadcast, and reconstruction—and uses only addition, subtraction, shift, and comparison.

Election. Every cell owns a predefined value $\nu \in [0,\, 9L-1]$—its W-island index, fixed once by the internal address—and at the expansion limit of the pulse (the exact tick where the wave turns from ascending to descending) every active shell cell is seeded with the payload

$$\mathrm{payload}(x) = \big(\mathrm{score}(x) \ll 24\big)\,\big|\,\mathrm{code}(x), \qquad \mathrm{score} = H\big((\nu(x)+s) \bmod 9L,\ \mathrm{code}(x)\big),$$

where $\mathrm{code}(x)$ is the cell's linear index, H a fixed integer hash, and s a global seed dial. Because the code occupies the low bits, the payload order is a strict total order: ties are impossible, so the winner is unique and independent of any scan order. Each pass lets every cell adopt the greatest payload among its six face neighbours,

$$\mathrm{payload}(x) \ \leftarrow\ \max\big(\mathrm{payload}(x),\ \mathrm{payload}(x \pm e_i)\big),$$

with alternating sweep directions; $2R+8$ passes guarantee full coverage of the lattice. The true winner—the *seeded* cell holding the greatest payload—is identified once, at sowing time, while the payloads are still unique; the diffusion flood merely transports the news and never redefines the winner. The displacement from the lattice center to the winning cell becomes the momentum vector $\boldsymbol{m}$, of which only the direction is meaningful: it is rescaled to $|\boldsymbol{m}| = L/2$ and frozen while the helical walk lives. The next expansion limit elects a fresh direction for the next run; nothing is prescribed, and the orientation is read off the automaton's own ledger at the instant the wave reaches its limit.

Broadcast. As soon as $\boldsymbol{m}$ exists, a helical walker traces the spiral arm on the pulsating bubble. It advances one cell per tick around the cylinder whose axis is the elected direction, keeping the radial error within about one cell by purely local comparisons among its six neighbours. The walker is the arbitrary-axis integer helix ported from `spiral/spiral.c`: at election time it computes the constant cross products $\boldsymbol{D}_m = \boldsymbol{e}_m \times \boldsymbol{m}$, the shift tables $K_{km} = 2\,\boldsymbol{D}_k \cdot \boldsymbol{D}_m$, the target radius $T_{\mathrm{GT}} = R_{\mathrm{CYL}}^2\,|\boldsymbol{m}|^2$ and the tolerance band $T_{\mathrm{OL}} = (2R_{\mathrm{CYL}}+1)\,|\boldsymbol{m}|^2$. The cylinder radius is $R_{\mathrm{CYL}} = (93\,R_{\max} + 256)/512$. During the walk the squared radial error E, the direction filters G_m, and the orbital displacement $\boldsymbol{v}$ (with $q^2 = |\boldsymbol{v}|^2$) are updated incrementally by addition, subtraction and shifts; an orbital step is chosen in the six-neighbour direction that keeps turning around $\boldsymbol{m}$ while staying inside the tolerance band, and a climb step along $\boldsymbol{m}$ is distributed by Bresenham so the walker also advances along the axis.

Each newly computed spiral point writes one plus the current tick into its own cell of a second value lattice, reserving $b(x) = 0$ for "never reached". Every other tick, with alternating sweep directions, every cell inspects only its six face neighbours and copies the greatest neighbour value whenever it exceeds its own,

$$b(x) \ \leftarrow\ \max\big(b(x),\ b(x \pm e_i)\big).$$

No cell ever senses beyond its neighborhood, yet these monotonic values diffuse as waves that in finite time reach every cell of the lattice—global coverage achieved purely by local moves, in lockstep with the automaton tick. Each cell therefore ends up holding the tick $b(x)$ at which the news of the elected momentum arrived.

Reconstruction. The arrival stamp is the phase source. A full 2π turn is encoded by $\mathrm{phase\_full} = 2R^2$ units, where $R = L/2 - 2$ is the emergent bubble radius, and the cell phase is read directly from the broadcasted value,

$$\mathrm{cell\_phase} = (b(x)-1) \bmod \mathrm{phase\_full}, \qquad j = \left\lfloor \frac{\mathrm{cell\_phase}}{R} \right\rfloor.$$

Writing $j_2 = j - R$ for $j \geq R$, the integer pair is reconstructed by

$$\text{if } j < R: \quad u = R(R-2j), \qquad v = 2R\,\mathrm{isqrt}\big(j(R-j)\big), \tag{4}$$

$$\text{if } j \geq R: \quad u = R(2j-3R), \qquad v = -2R\,\mathrm{isqrt}\big(j_2(R-j_2)\big). \tag{5}$$

These formulas generate an integer pair (u,v) that lies on the integer lattice closest to the circle $u^2 + v^2 = R^4$ of radius R^2 (the equality is exact when the integer square root is exact, and is otherwise the closest integer approximation). The polarization angle is $\theta = \mathrm{atan2}(v,u)$; it advances from 0 to 2π as the broadcasted value grows, winding the polarization helix around the elected momentum axis. No trigonometric tables, no precomputed spiral, and no per-cell fixed constants are used: the direction emerges from the tournament over the W-ledger, and the phase from the integer square-root relation applied to the arrival time.

Once these fields are present, the cyclic evolution described in Sect. 6 begins. Within that structured loop particles move, interact, and display emergent physical behavior.

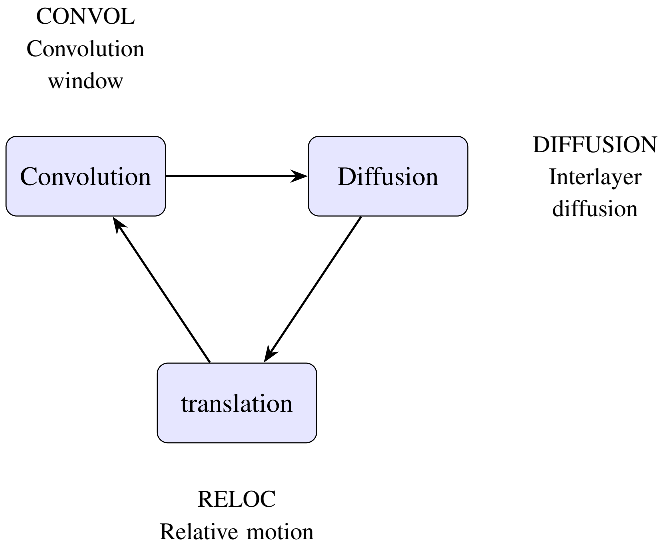

FIGURE 2
The light frame: Convolution, Diffusion and translation. Constants CONVOL, DIFFUSION and RELOC delimit the phases.

6 THE LIGHT FRAME

The light frame (Figure 2) is one complete light step of the cellular automaton. It is divided into five stages: `CONVOL` (interactions and pair formation), `DIFFUSION` (propagation of contact, collapse, and translation information), `RELOC` (physical displacement of source centers), `REISSUE` (re-emission of the wavefront from the new source center), and `FLOOD` (final synchronization of the three lattices). The global housekeeping counter k advances synchronously in every cell, marking the current stage; the wavefront counter t is local and resets to zero whenever a bubble is reissued. The separation between the common k and the local t is what lets the finite-state machine coordinate seemingly non-local operations without introducing any superluminal signal.

6.1 Definitions

The light-frame constants are:

- **DIAG**
  $\lfloor\sqrt{3}\,L\rfloor$: diagonal length of the lattice; upper bound for interaction distances.
- **RMAX**
  $DIAG/2$: maximum bubble radius before wrapping.
- **CONVOL**
  $2W$: convolution window.
- **DIFFUSION**
  $CONVOL + 2W + \lfloor\sqrt{3}(L-1)/2\rfloor + 9(L-1) + 2W$: diffusion range along W; also alerts all layers of an imminent collapse.
- **RELOC**
  $DIFFUSION + 3(L-1)$: translation phase.
- **REISSUE**
  $RELOC + 1$: re-emission of the wavefront from the new source center.
- **FLOOD**
  $REISSUE + 3(L-1)$: final synchronization of the three lattices.

- **FRAME**
  Same value as $FLOOD$; total duration of the light step.

6.2 Bubble, pair, singleton, and blob

A *bubble* is an expanding spherical wavefront of information organized across cells within a common layer. Spherical propagation is determined by comparing the dynamic radius r to the wavefront tick t.

Pair
Two bubbles are overlapping if they share the same spatial coordinates and $t_1 = t_2$. Overlapping bubbles can form a *pair* if they have the same affinity, that is, $a_1 = a_2$; the two partners share a center and phase, and their charges are complementary or matching depending on the pair's role. Pairs may remain bound to a particle as *dressing* pairs or propagate freely as photons; a free pair expands outward and is gradually consumed, releasing two singletons at distinct locations.

Propeller
A pair can act as a *propeller*, repositioning other bubbles and providing inertia.

Singleton
An unpaired bubble is called a *singleton*; it is not a generic synonym for bubble.

Blob
The *blob* is a group of superposed pairs having the same affinity, same true bB bits and the same (active) pBits.

Specialized pairs
Certain pair combinations are not allowed to form blobs:

- Neutrino and antineutrino pairs possess color values N, N and $\bar{N}, \bar{N}$, respectively.
- The up quark fragments u exhibit equal, nontrivial colors and a positive charge in the case of Orbis.
- The propeller adds to the relativistic mass of the propelled fermion.†
- Pairs with fully complementary charges are *gravitons*. They operate in both Orbis and Umbra,‡ have fixed oscillation frequency 2, and mediate static cross-sector interactions; without them, Umbra matter would appear inert in Orbis. Graviton-induced gravity is detailed in Sect. 12.4.

6.3 Interaction principles

The guiding principles are:

Wavefronts clash
The wavefronts of both bubbles must be active, i.e. $r_1 = \text{effective\_t}(t_1)$ and $r_2 = \text{effective\_t}(t_2)$, where r_i is the propagated integer radius of bubble i and t_i is its local light-frame counter. This condition is assumed in all rules below.

Aggregation and dissolution
Bubbles aggregate by sharing a common affinity; the smallest value prevails, $a_1 = a_2 = \min(a_1, a_2)$. On dissolution, they revert to their layer indices, $a_1 = w_1$ and $a_2 = w_2$. If a singleton meets a pair, the pair takes the singleton's affinity, $a_2^1 = a_2^2 = a_1$. The value $a = W$ marks an orphan. The idempotent $\min$ operator naturally supports an ontological model of entanglement.

Charge conjugation
Charge conjugation constrains interactions by requiring charge neutrality in the outcome. Electric neutrality requires a pair of opposite charges; a single bit cannot be neutral alone. Weak neutrality follows from the combination of the two weak bits in both *Orbis* and *Umbra*. Strong neutrality is 000 (or 111 for anti-neutrality); a pair with complementary color bits (e.g. 101 and 010) is also neutral. In general, interactions drive the resulting charge combination toward neutrality.

† This mirrors a Kantian view: particles do not self-move but require an external cause—here, interaction with fragments—to change their state.

‡ Orbis and Umbra coexist spatially.

Reciprocity
Because only one draft can be updated at a time, the rules must later update the counterpart. This reciprocity is essential for the next subsection.

6.4 Convolution

Convolution is the first step of the light frame (ending at $CONVOL$) and contains the main interaction rules. Each site in the main lattice is compared with its mirror along the W dimension. During this step, the wave phase f advances to select the wave-cloud point density, so it also embodies the notion of frequency. Particles form or reform here. We postulate that the weak force only causes collapse, not limited reissues. Comparisons split into two cases: superposing or distinct bubbles.

Superposing bubbles
Two overlapping bubbles with $f = t$ can form a pair if they satisfy the following conditions:

- $(q_a \oplus q_b) \wedge (w1_a \oplus w1_b) \wedge (w0_a \oplus w0_b) \wedge (c2_a \oplus c2_b) \wedge (c1_a \oplus c1_b) \wedge (c0_a \oplus c0_b)$
- $(q_a \oplus q_b) \wedge (w1_a = w1_b) \wedge (w0_a \oplus w0_b) \wedge (c2_a \oplus c2_b) \wedge (c1_a \oplus c1_b) \wedge (c0_a \oplus c0_b)$
- $ch_a = ch_b = 000000\text{B}$
- $ch_a = ch_b = 111111\text{B}$
- $(q_a = q_b = 0) \wedge (w1_a = w1_b = 0) \wedge (w0_a = w0_b = 1) \wedge (color_a = color_b) \wedge (color_a \neq 000\text{B} \wedge color_a \neq 111\text{B})$
- $(q_a = q_b = 1) \wedge (w1_a = w1_b = 1) \wedge (w0_a = w0_b = 0) \wedge (color_a = color_b) \wedge (color_a \neq 000\text{B} \wedge color_a \neq 111\text{B})$

These rules implement the formation of pairs, as defined in Sect. 6.2. If pairs are detected, the wave phase advances by $f \leftarrow f + t$. $s2B$ is updated as $s2B' = s2B$ **and** $active$. Notice that the pairs are formed in two steps: first, cell a compares with cell b, and later, cell b compares with cell a (reciprocity). The formation of blobs is made during this step too, also following the reciprocity principle.

Pairs are allowed to superpose if they are equal and were formed by the two first rules above.

Distinct bubbles
The w_1 bit separates distinct-bubble interactions into same-sector ($w_1^1 = w_1^2$) and inter-sector cases; we first treat same-sector interactions.

If two same-sector singletons have opposite charges and both kB bits false, we have annihilation, with both bubbles being reissued from the contact point and their affinity receiving their default values. The draft kB bit turns true, triggering a collapse.

Else if two singletons have the same charge, we have fermion cohesion. One bubble is reissued from the contact point, while the other, from its active sB site. Also, they become entangled ($a_1 = a_2$).

On the other hand, if the two bubbles share the same affinity, but have different charges, the bubble whose wavefront at the contact point has $pB = 1$ is reissued from the contact point. The other bubble is reissued using the inertial-transported translation impulse, accumulated in the vector $\boldsymbol{reloc}$; the stable momentum vector $\boldsymbol{m}$ supplies the direction of that impulse, while the translation vector $\boldsymbol{c}$ is consumed during the reissue step. That is, one bubble is reissued from $\boldsymbol{x1}$, while the other from $(\boldsymbol{x_1} - \boldsymbol{x_2})\, \boldsymbol{mod}\, L$.

For color-neutral combinations there are two cases: gluon×gluon and quark×gluon. In both cases, the two bubbles are reissued from the contact point and the partners colors are exchanged.

In the electroweak sector, both $active$ bits must be true; together with $s2B$ they determine the harmonic behavior. This sieve mechanism applies only to electroweak interactions and helps explain why the strong force is stronger.

The weak interaction occurs when the weak charges are combined to neutrality and either ($pB_1 = 1, pB_2 = 0$) or ($sB_1 = sB_2 = 1$); it always triggers collapse ($kB = 1$).

The electric case uses one pB bit, the magnetic case one sB bit. If both pB bits are true (or both sB bits in the magnetic case), a collapse is triggered ($kB = 1$).

If only one pB bit meets the other's surface, the interaction is adiabatic: no collapse occurs, but the bubbles exchange affinity and time and relocate to each other's center. The a and t values then spread throughout the layer.

At the initial singularity, all bubbles overlap. If two bubbles belong to different sectors and $t = RMAX/2$, they are reissued from their active pB and sB sites, dispersing the singularity and separating Orbis from Umbra. We call this *dispersion*.

Inter-sector interactions
For inter-sector interactions, fully complementary charges lead to *singularization*; otherwise the electroweak rules above apply. Singularization is the counterpart to dispersion. The bubbles are reissued from the contact point (pB_{curr} = true).

6.5 Diffusion

Diffusion propagates the information needed for translation, ending at the `DIFFUSION` tick (Figure 3). If any collapse bit kB is set, it spreads to all bubbles sharing the same affinity, identifying every contributor to the collapsing particle.

The wave phase f propagates as

$$f_{\text{draft}} = \max(f_1, f_2),$$

governing photons and bosons. Phases incremented by multiples of the fundamental cycle interact with the wave-cloud sieve to form concentric bubbles; the active wavefront point ($t = d$) is forwarded to the Euclidean bubble through $s2B$.

At the same time, the translation vector **c** diffuses through the layer: each cell copies the first non-trivial **c** found among its neighbors. Finally, sB hunting increments $\boldsymbol{c}$ along any axis where a neighbor has $hB = 1$, but only when the local cell has $sB = 0$. By the end of diffusion, $\boldsymbol{c}$ is ready for translation.

6.6 translation

This part of the light step addresses bubble translation itself, building on the diffusion step. It ends at the RELOC tick. Bubbles shift in specific directions (north, west, down) until they have completed their translation process, including the inertial transport mechanism. The example rule is: if $\mathbf{c}_x^{north} > 0$, copy the content of the neighbor cell to the draft cell and decrement $\mathbf{c}_x$ at the draft cell.

Since wavefront propagation is guided by the dynamic radius r propagated from the source cell, when a bubble is re-emitted due to interaction with other bubbles, all pertinent cell information must be shifted, or relocated, to the new position determined by the variable **c**. This ensures that the bubbles are relatively displaced from each other. The final operation sets $t = 0$ at the center cell of the relocated pattern. translation is the main consequence of an interaction.

Bubbles are re-emitted either immediately after an interaction or through wrapping. When the bubble is reissued, the previous wavefront continues to propagate until it fades through wrapping. In this state, the bubble is referred to as an *orphan*, meaning it has affinity disabled but is still capable of non-collapsing interactions. In this case, $a = W$ for the outer cells, set during the translation step.

To complete, the diffusion of the time variable t occurs here at the last tick: When a cell has a neighboring cell with a lower t value, it updates its own t variable to match the lower value. With this rule, the relocated bubble is capable of expanding again. This timing is crucial to avoid the 'clean diamond' pattern with the new values overrunning the orphaned data.

When the translation is complete, bubble variables are prepared for the next cycle. If the time frame completes, the k counter resets to zero, preparing the system for the next iteration. At the conclusion of the process, all $\boldsymbol{c}$ vectors will have a null value.

A propeller (P) can only be relocated to a point on its active surface. In a $P \times K$, $P \times D$ or $P \times S$ interaction, the P source is reissued from the active contact cell where the two wavefronts overlap; this contact point is already reached at the speed of light, so no superluminal signal is required.

Inertia is acquired by an inertial-transport mechanism. The propeller transfers its stable momentum direction $\boldsymbol{m}$ to the partner's consumable translation impulse $\boldsymbol{reloc}$; during the next light step the partner's source center is shifted by $\boldsymbol{reloc}$, while its long-term $\boldsymbol{m}$ is preserved and updated by `applyMomentum()`. The P source itself is reissued from the contact point; the partner (K, D or S) keeps its source kind but receives the impulse.

Non-propeller pairs behave differently: in $K \times K$ or $D \times D$ between different tribes the source centers move one light-step away from each other, exchanging momentum through $\boldsymbol{reloc}$.

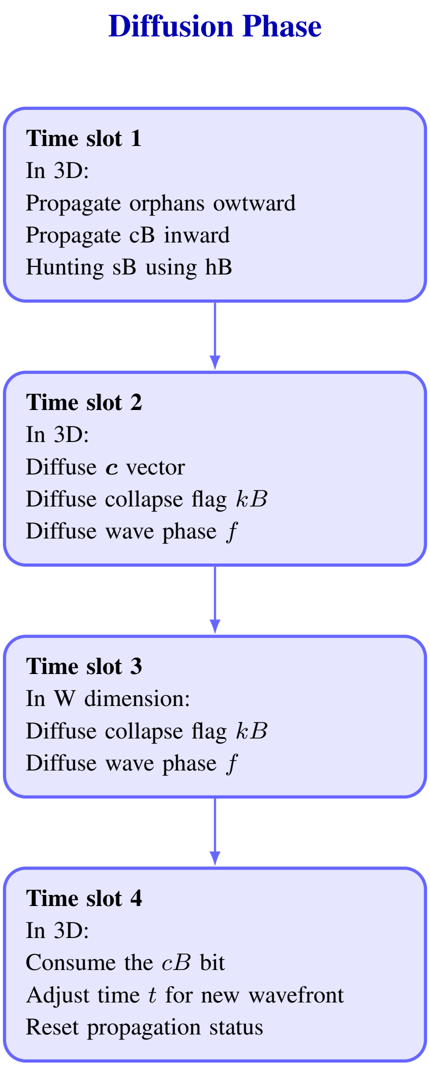

FIGURE 3
The Diffusion phase is composed of several time slots. It spreads all information necessary for translation.

The propeller (Sect. 6.2) is a pair (P) that acts as a carrier of momentum. Free photon or graviton pairs may become propellers through the same P source mechanism, but during interaction the source kind P is treated as a single relocating source whose momentum is transferred to the non-P partner.

6.7 Annihilation and collapse

Annihilation and light-matter collapse both occur during convolution. In annihilation, the affinity reverts to its default value $a = w$, and the resulting bubbles become raw material for new particles.

This ontological collapse introduces an arrow of time at the fundamental level, reinforcing the second law of thermodynamics and pushing Poincaré recurrence far into the future (Sect. 13.1).

6.8 Main lattice updating and mirroring

At the end of each k tick, the draft lattice is transferred to main. If k lies inside the $CONVOL$ window, draft is first shifted along W to enable convolution. The mirror lattice is then refreshed from main so the next convolution can compare main and mirror along W; it copies the wave phase and active flag, $f_i^{mirror} = f_i^{main}$ and $active_i^{mirror} = active_i^{main}$.

7 INTERACTION DETAILS

Some interaction details are now clarified.

7.1 Charge driven interactions

Static interactions

A static interaction (Coulomb or magnetic) between two distant charges q_1 and q_2 begins when an orphan singleton from q_1 aligns with a vacuum graviton; both are reissued at the contact point. If the graviton then interacts with q_2, it is reissued there and acquires q_2's affinity. Thus gravitons propagate along the flux of orphans from particle 1, increasing the chance of hitting q_2 and becoming one of its propellers. The same occurs in reverse.

Coulomb interaction

Coulomb attraction and repulsion occur in two steps: (a) the wavefront of an orphan from the first charge, with $pB = 1$, contacts an active cell of a graviton; if their position vectors are aligned or anti-aligned, the graviton is reissued from the contact point with the orphan's affinity; (b) the second charge's wavefront, also with $pB = 1$, contacts the reissued graviton and receives a translation impulse in the direction of its momentum vector $\boldsymbol{m}$, becoming a propeller pointing in the correct direction.

Static magnetic interaction

The magnetic case uses sB instead of pB and applies to charged leptons, quarks and W bosons. Because propellers emphasize pB over sB, fast particles tend to acquire a component perpendicular to their direction of motion.

Neutrino interactions

Let $q_{0,1}$ be a complementary electric-bit pair. A neutrino fragment has charge content $q_{0,1} = 00000B$; an antineutrino has $q_{0,1} = 01111B$. (Anti)neutrino fragments interact only with (right-)left-handed particles.

The up quark and down quark fragments

The color combinations of fragment pairs are shown in Fig. 4. (Anti)*quark* fragments have non-trivial color and are marked as q or $\bar{q}$, (anti)neutrinos are ν and $\bar{\nu}$, gluons are g, and photons are γ. Since half the gluon fragments are right-handed, they are shown as photons. The cells marked as blue, if having both quarks with electrical charge $q = +$, may form an *up quark* fragment.

An *up quark* u can be any of the *+R+R*, *+G+G* or *+B+B* pairs, while a *down quark* d can be any of the *-R*, *-G* or *-B* single fragments. A proton fragment *uud* has an uncompensated color, balanced dynamically by the matter and antimatter ends of gluons. Hence the *electron* fragment must be $-N, -N, -N$.¶

The number of up-quark fragments in each sector sets an upper limit on proton, and hence *atom*, production.

¶ This explains fractional quark charges.

	N	R	G	$\bar{\text{B}}$	B	$\bar{\text{G}}$	$\bar{\text{R}}$	$\bar{\text{N}}$
	000	001	010	011	100	101	110	111
N 000	ν	1q 1l	1q 1l	$1\bar{q}$ 1l	1q 1l	$1\bar{q}$ 1l	$1\bar{q}$ 1l	γ
R 001	1l 1q	2q	2q	γ	2q	γ	γ	$1\bar{l}$ 1q
G 010	1l 1q	2q	2q	g	2q	γ	γ	$1\bar{l}$ 1q
$\bar{\text{B}}$ 011	1l $1\bar{q}$	g	g	$2\bar{q}$	g	$2\bar{q}$	$2\bar{q}$	$1\bar{l}$ $1\bar{q}$
B 100	1l 1q	2q	2q	g	2q	γ	γ	$1\bar{l}$ 1q
$\bar{\text{G}}$ 101	1l $1\bar{q}$	g	g	$2\bar{q}$	γ	$2\bar{q}$	$2\bar{q}$	$1\bar{l}$ $1\bar{q}$
$\bar{\text{R}}$ 110	1l $1\bar{q}$	g	g	$2\bar{q}$	γ	$2\bar{q}$	$2\bar{q}$	$1\bar{l}$ $1\bar{q}$
$\bar{\text{N}}$ 111	γ	1q $1\bar{l}$	1q $1\bar{l}$	$1\bar{q}$ $1\bar{l}$	1q $1\bar{l}$	$1\bar{q}$ $1\bar{l}$	$1\bar{q}$ $1\bar{l}$	$\bar{\nu}$

FIGURE 4
Color combination.

Possible combinations of colored fragments are shown, where (anti)quark fragments are marked q and $\bar{q}$, (anti)neutrinos are ν and $\bar{\nu}$, gluons are g, and photons are γ. Since half the gluon fragments are right handed, they are shown as photons. The cells marked as blue, if having electric charge $q = +$, may form an *up quark* fragment in Orbis.

7.2 Self-interference

Space in this model retains a memory of particle trajectories, producing self-interference reminiscent of the double-slit experiment [12]. The trace is encoded by the last affinity value a and the most recent wave phase f left in visited cells.

During convolution (Sect. 6.4), incoming and residual bubbles are compared. If $d = t$, standard cohesion occurs; if $d = f$, only the new bubble is reissued, altering the fermion's dynamics and creating interference patterns. This temporally shifted interaction among the constituent "bits" of qubits (Sect. 12.19) provides a substrate for complex quantum behavior in a discrete framework.

8 PARTICLES

Particles are composite structures of bubbles. A particle is primarily identified by most of its bubbles sharing the same affinity a. Its stability comes from dynamic charge quantization: the 3D island population emerges from a balance between bubble capture and escape, while surrounding pairs act as propellers for transport.

In a complementary pair, weak and strong interactions are inhibited (but electric charge remains active). Both bubbles are reissued in a pair interaction, and such combinations generally form boson fragments.

8.1 Fermions

A fermion contains more than paired kinematic components. It typically has a 3D island sustained by dynamic charge quantization (Sect. 9)—the up quark is the specialized pair of Sect. 6.2—plus propeller pairs responsible for inertial motion. Fermions form a small, "punctiform" bubble cloud surrounded by concentric orphans from frequent reissues, which constitute their *field*.

Quark confinement follows from the priority of color cohesion and quark-gluon interaction, well different from the electromagnetic case. In atoms, electrons distribute around the nucleus according to spherical harmonics.

Specialized pairs do not form blobs and therefore have fermionic character; the neutrino pair is an example. Inter-sector neutrino pairs do not interact via the weak force, making them effectively sterile. Neutrino mass is the sum of the pair components plus propeller contributions.

8.2 Bosons

A boson is made of one or more blobs. Photon fragments and other gauge-boson fragments are examples; their pairs differ in some or all charge bits except w_1, making them sectorial particles.

A photon is a single blob with all pairs superposed. In this model the frequency of a photon is not a continuous parameter but a count: a single overlapping pair at the same lattice site and wavefront radius has the fundamental even frequency 2; a stack of n such pairs therefore has frequency $2n$. A multifrequency photon is a superposition of several such stacks, or a blob containing populations at different even frequencies $2, 4, 6, \ldots$. There are no harmonics in the continuous sense; the spectrum is discrete and composed only of these even-integer modes.[§]

$W^{\pm}$ and Z bosons, by contrast, form a "punctiform" cloud of several blobs sharing the same affinity, somewhat like fermions. The Higgs boson is conjectured to be two Z^0 bosons with opposite spins, with no apparent relation to giving mass to other particles.

Orphans ($a = W$) do not interact with photons but do interact with gravitons. Neutral orphan pairs do not interact electromagnetically.

9 DYNAMIC CHARGE QUANTIZATION AND THE SCALING OF *W*

This section expands Sect. 4.1 and details the dynamic organization and quantization of W-islands.

9.1 Objects and identity

A *3D island* is a spatially localized collective of singletons with the same dominant charge and affinity. A complete *particle* is a 3D island of bare charge plus a surrounding *pair dressing*:

$$\text{particle} = \text{3D island} + \text{pair dressing}. \tag{6}$$

The internal non-spatial dimension is partitioned into

$$N_I = 9L \tag{7}$$

identical *W-islands*, each containing

$$\ell = \frac{L}{3} \tag{8}$$

positions. A W-island is not an object in physical space; it is an internal charge family. All ℓ copies inside one island share the same six charge bits (q, w_1, w_0, c_2, c_1, c_0), but each copy carries a distinct pB/sB pattern. This built-in multiplicity is why identical particles are so similar, while the distinct polarizations allow the radial Hofer pattern (Sect. 12.9) to emerge.

Initially the $L/3$ copies of a W-island are singletons (S). When two copies of the same W-island meet in the 3-torus, one becomes the chief (K) and the other a delegate (D). Additional copies that later meet this island are captured as further delegates. A mature 3D island therefore has the form

$$1K + nD. \tag{9}$$

Copies of the same W-island always merge into a single $1K + nD$ island; distinct W-islands interact through the usual $K \times K$, $D \times D$, and convolution rules. The chief is thus not selected by a modulo operation or by a fixed index during initialization; it emerges dynamically among the identical copies. Each bubble keeps its intrinsic W address w immutable; only the auxiliary identity `leader_w`, copied from the chief, converges to the dominant value after mergers. The total number of possible chiefs is bounded by $9L$, and the total primitive bubble population is

$$3L^3 = (9L)\left(\frac{L}{3}\right). \tag{10}$$

[§] Laboratory and environmental photons are groups of such blobs, combined in time and space.

9.2 Dynamic charge quantization

A 3D island is an open dynamical object: it continuously captures free S bubbles and loses surface bubbles. Let $\Gamma_{\text{cap}}(N)$ and $\Gamma_{\text{esc}}(N)$ be the effective capture and escape rates for an island of characteristic population N. Dynamic charge quantization is the statement that

$$\Gamma_{\text{cap}}(N) > \Gamma_{\text{esc}}(N) \quad \text{for small } N, \tag{11}$$

$$\Gamma_{\text{cap}}(N) < \Gamma_{\text{esc}}(N) \quad \text{for large } N. \tag{12}$$

A stable attractor N_* then appears near

$$\Gamma_{\text{cap}}(N_*) \simeq \Gamma_{\text{esc}}(N_*), \tag{13}$$

without any global counter. The characteristic charge of a particle is the statistically preferred population of this attractor, dressed by the required photon pairs.

9.3 Origin of W and FSM addressing

The non-spatial dimension has size

$$W = 3L^2. \tag{14}$$

The factorization

$$W = (9L)\left(\frac{L}{3}\right) \tag{15}$$

splits the ledger into $9L$ charge families, each with $L/3$ identical copies. In a finite-state machine implementation the W coordinate is therefore not a single integer accessed by modulo; it is a pair of registers, an island index in $[0, 9L)$ and a copy index in $[0, L/3)$. This preserves the model's FSM character: all addressing and state transitions remain local and involve only comparison, copying, and small counters.

At this point, the core construction of the cellular automaton is complete.

10 RESULTS

We now give a few practical examples.

10.1 The speed of light and the size of the universe

We can now state the following constraint on the fundamental quantities:

$$\frac{X}{\mathcal{L}} = c,$$

where c is the speed of light and $\mathcal{L}$ is the number of ticks per frame. Estimates for the values of *X* and *L* (truncated to an integer) are taken from Appendix A

$$X = l_P$$

and

$$L \approx 6.95410 \times 10^{60} \approx 2^{202},$$

where l_P is the Planck length. From this, we calculated the size of the universe as $1.94669 \times 10^{26}\, m$ (the *Grand Cube* diagonal).

10.2 A charge conjugation study

A small program (Ref. [13], *combine.c*) combined $6 \times 32768 = 196{,}608$ charge-balanced fragments in several million random attempts, tabulating the averages and assuming only "hydrogen atoms" form.

Since the strong force dominates, the search used probability 0.001 for up quarks and 0.999 for gluons, based on the smallest possible charge set. The search proceeded in four stages: gluons and up quarks (plus electrons, whose number depends on up quarks), then photons and neutrinos, then W and Z fragments, and finally anti-atoms.

Results are in Table 2. Leftover bubbles that cannot form a pair or singleton indicate an ionized environment; inter-sector attenuation increases the photon count. The data roughly resemble a few simple atoms plus multiple-pair photons. Unmatched quark fragments may become mesons, virtual quarks, or contribute to dark matter (50%

TABLE 2
Charge combination

Fragment	Type	Orbis	Umbra	Obs.
Gluon	Pair	3,6760.0	3,6760.0	
Up quark	Pair	42.0	42.0	
Down quark	Single	10.0	10.0	
Electron	Single	31.0	31.0	
Photon	Pair	20.0	20.0	
Graviton	Pair	12.0	12.0	
Z boson	Pair	24,592.0	18,416.0	
W boson	Pair	6,122.0	12,294.0	
Neutrino	Pair	6,082.0	6,082.0	
Antiup	Pair	2.0	2.0	
Antielectron	Single	1.0	0.0	
Antiquark	Single	0.0	0.0	
Leftover	Single	24,624.0	24,620.0	Adds to dark matter
Total		**196,602.0**		

A total of 196,608 bubbles were randomly combined into several million attempts in a small program focused on charges alone (See Sect. 10 for more details).

+ 25%); Umbra appears to contain more antimatter than Orbis. Despite its naivety, the outcome is sensible as a toy universe.

The proton-electron mass ratio comes out as 1187 in both sectors, somewhat close to the empirical value ≈ 1836 (CODATA [14]).

10.3 Full implementation in a small lattice

A full implementation of the rules is provided in Ref. [13], including source code and an executable. The lattice size is limited to $L = 11$ by available computational resources, but this still serves as a proof of concept for visualization and verification.

The next section discusses remaining challenges and limitations.

11 CELLULAR AUTOMATA AND QUANTUM FORMALISM

For local events, we follow 't Hooft's cellular-automaton world-model approach [9]. Ontological CA states ($\mathcal{OS}$) do not superpose: each evolves into another ontological state. An orthonormal set of ontological states can nevertheless serve as a Hilbert-space basis.

The general operator-mechanics recipe is:

- Find a permutation operator as a matrix containing a single 1 in any row or column.
- Diagonalize it, obtaining the eigenvalues and eigenvector matrices.
- Identify a Hamiltonian matrix as the exponent in $\hat{U} = e^{-i\hat{H}T}$.

- Use the eigenvalue matrix to bring this Hamiltonian back to the ontological basis.
- Form linear combinations of the ontological states $|A\rangle$.

$$|\psi\rangle = \sum_A \lambda_A |A\rangle\,, \quad \sum_A |\lambda_A|^2 \equiv 1,$$

  where $|\psi\rangle$ is a quantum state or "template," and λ_A is a complex coefficient.

With this operator machinery, the Schrödinger equation governs the deterministic evolution of the wave function:

$$\frac{d}{dt}|\psi\rangle = -i\hat{H}|\psi\rangle\,.$$

The Born rule follows once superposition states are accepted; it predicts the final eigenstate after measurement. This is a simplified picture; details are in [9, 15, 16].

12 CONJECTURES AND PROSPECTS

We briefly discuss some open conjectures.

12.1 Discrete to continuous transition

Localized particles are assemblies of many discrete fragments, so macroscopic properties such as total momentum and angular momentum emerge from collective configurations rather than from individual bubbles. These global quantities are therefore best described by real numbers, just as statistical mechanics bridges discrete atomic events to continuous thermodynamic variables.

A free electron's average motion, for example, may follow a direction vector that cannot be expressed with only discrete modulo-L components; continuous descriptors are thus necessary idealizations at large scales. The same holds for isotropy: it looks uniform at macroscopic scales even though the lattice is fundamentally granular.

12.2 Color quantization

Color quantization is an extension of the electromagnetic case. Hadrons form by the simultaneous action of charge quantization and strong cohesion, producing color-neutral bound states of quark and gluon fragments. In the discrete color code (Sect. 6.2), a strongly neutral combination is either a pair of complementary color bits (meson-like), a pair of identical non-trivial colors (diquark-like), or a triplet of the three distinct colors (baryon-like, e.g. a proton uud fragment). Free quarks are not observed because only these color-neutral composites are dynamically stable; any uncompensated color is rapidly balanced by quark-gluon exchange.

12.3 Special relativity

Time intervals, spatial separations and clock synchronization follow Einstein's special relativity [17]: time is read by ideal clocks, distances by round-trip light-travel times, and coordinate time by a network of synchronized clocks.

The coordinate W can be seen as a fourth spatial dimension, where the now 4D-bubbles form a thin shell, to avoid known 4D mutual passing issues (see Machotka [18]). In this case, using real quantities as an approximation, we can write a metric with signature $(+,+,+,+)$:

$$c\,\delta t = \sqrt{\delta x^2 + \delta y^2 + \delta z^2 + \delta w^2}, \tag{16}$$

where δx, δy, δz, and δw are infinitesimal increments, and δt is the infinitesimal increment of coordinate time.

The magnitude of the 4D velocity is

$$v_{4D} = \sqrt{v_x^2 + v_y^2 + v_z^2 + v_w^2} = \sqrt{v_{3D}^2 + v_w^2}, \tag{17}$$

where v_x, v_y, v_z, and v_w are components of the 4D velocity vector (in 4D space) while v_{3D} is the standard three-dimensional velocity (in ordinary 3D space).

Substituting $v_{3D} = \sqrt{\delta x^2 + \delta y^2 + \delta z^2}/\delta t$ into Equation (16) gives

$$c^2\,\delta\tau^2 = c^2\,\delta t^2 - v_{3D}^2\,\delta t^2. \tag{18}$$

Thus,

$$\delta\tau = \sqrt{1 - \frac{v_{3D}^2}{c^2}}\,\delta t, \tag{19}$$

which is the familiar time dilation formula from special relativity. In this way, we can claim that the model is Lorentz invariant.

12.4 Emergence of gravity

General-relativistic spacetime is an effective, coarse-grained approximation of the discrete lattice. Gravity is conjectured to arise as a residual effect of electromagnetic interactions, mediated by gravitons that couple to both *Orbis* and *Umbra*. Ordinary photons are sector-specific, but gravitons provide cross-sector connectivity; their electric and weak contributions largely cancel, leaving a residual geometry-driven attraction. Free objects then follow emergent timelike geodesics, and graviton redistribution of energy and charge may mimic dark-matter effects such as flat rotation curves.

Photons may also lose energy through minor, non-collapsing interactions with singletons and electrons as they propagate—an intrinsic *aging* that parallels cosmic redshift. Zwicky already pointed out a redshift mechanism independent of expansion [19]; the present model does not suffer from the image-blurring problem he raised. This raises the possibility of a redshift explanation that does not require dark energy. The propeller mechanism also makes acceleration equivalent across frames, in line with the equivalence principle.

12.5 Weak quantization

Because weak interactions are rare, weak quantization may appear at cosmological scales, for instance in a constant ratio between the average galaxy separation and galactic bulge size. This large-scale granularity could be reinterpreted as gravitational quantization and may eventually yield an effective cosmological constant Λ and metric tensor $g_{\mu\nu}$.

12.6 Stochastic behavior and the uncertainty principle

The initial chaos created by the deterministic collision of many regular patterns, together with the large number of bubbles per particle, makes individual outcomes effectively unpredictable. Each measurement samples one instance from this fine-grained distribution, so measured values carry irreducible uncertainty, analogous to the Heisenberg principle [20]:

$$\sigma_x \sigma_p \leq \frac{\hbar}{2}.$$

12.7 Black holes

The quintessential idea of a black hole in this model can be illustrated by two electrons, each with radius R, so close to each other $d < 2R$ that the quantization is broken and they simply fuse, forming a particle with radius $R' = 2^{1/3}R$ with a dominant affinity value.

$$a_1 = a_2 = \min(A_1, A_2).$$

In this picture, there is no black hole information paradox [21, 22].

12.8 What is spin in this model?

Spin in this model is the coherent radial alignment of the pBit pB (local in-phase sign) and the transverse polarization pair $(\mathrm{pol}_u, \mathrm{pol}_v)$. The positive in-phase half-cycle marked by pB alternates outward and inward as the wave oscillates, while the phase of $(\mathrm{pol}_u, \mathrm{pol}_v)$ winds once around the wavefront. This radial helix defines the two spin states, aligned consistently across the particle's constituent bubbles.

12.9 The Hofer effect

A key test is the *Hofer effect*: the predicted tendency of all individual spins of a localized particle to align radially inward or outward (spin down/up). Hofer [23] argues that magnetic effects arise from breaking the spherical symmetry of this pattern, offering an interpretation of the Stern–Gerlach experiment.

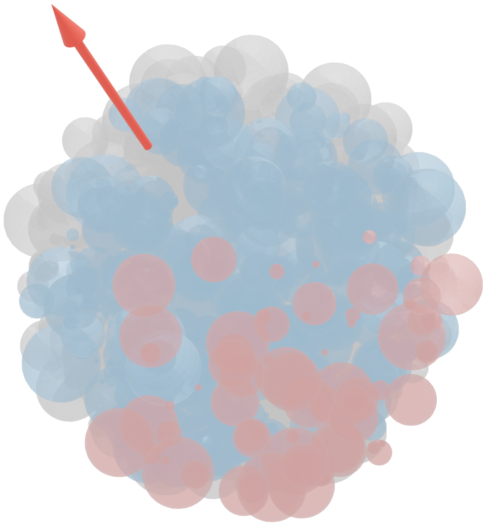

FIGURE 5
The free electron.

Artistic impression of a free electron. The red bubbles are propellers with a resultant in the direction of the arrow. The blue bubbles are negative charge singletons, while the gray bubbles are electron-neutrino pairs.

12.10 Particle zoo panorama

As it stands, this model serves as a framework to speculate on the composition of known particles. In the discussion that follows, the fragments will be characterized by their charge content: $q, w1, w0, c2, c1, c0$. For example, we designate the fragment 100000 as $-L$ and 001111 as $-\bar{L}$, and so forth. The following descriptions are expected to occur naturally due to the evolution of the system.

The electron neutrino
The electron neutrino ν_e is formed by a single $[+L : -L]$ pair and additional propellers. Yet the anti-electron neutrino $\bar{\nu}$ uses the pair $[+\bar{L} : -\bar{L}]$.

The electron
The electron e^- is formed by a quantized number of $-L$ fragments, a great number of electron neutrinos ν_e, and a variable number of propellers (see Figure 5). The positron, or anti-electron, uses $+\bar{L}$ fragments instead. These are for Orbis, in the case of Umbra, all bits are inverted.

The muon neutrino
The muon neutrino ν_μ is formed as $\nu_\mu = 2\nu_e + \bar{\nu}_e$ plus additional propellers. It is like an electron neutrino, but with a compensated antimatter part.

The muon
The muon μ^- consists of a quantized number of $-L$ fragments, a large number of muon neutrinos ν_μ, and a variable number of propellers. The classical study conducted by Itô in [24] suggests that this configuration stabilizes at the second harmonic of the electron's radial vibrational state.

The tau neutrino
The tau neutrino ν_τ is formed as $\nu_\tau = 3\nu_e + 2\bar{\nu}_e$ plus additional propellers. It is also like an electron neutrino, but with a compensated antimatter-enriched part.

The tau

The tau τ^- is formed by a quantized number of $-L$ fragments, a great number of tau neutrinos ν_μ, and a variable number of propellers.

Gauge bosons

These bosons were described in Sect. 8.2.

12.11 Leptons decay

In the Standard Model (SM), the decay products typically include one neutrino or antineutrino. However, in our model, the large number of neutrinos is entangled, sharing a common affinity. When one neutrino interacts via the weak force, all other entangled neutrinos collapse to the same point. This phenomenon gives the appearance of a single, energetic neutrino, as the observable outcome remains indistinguishable, in agreement with the SM predictions. If the neutrino belongs to the other sector (Orbis/Umbra), then we have spontaneous decay.

12.12 Electronic decay

After the absorption of (part of) a photon, the electron enters an excited state, dynamically stabilizing in a suitable spherical harmonic configuration. When the common components of the photon and electron are reissued, the odds of a configuration corresponding to the original photon are less than one. As time passes, after many cohesion interactions, the proper configuration of charges and pBits may occur, forming a single photonic blob and allowing the emitted photon to escape the frequent reissue mechanism, fleeing away from the electron cloud, which eventually settles into its ground state.

12.13 Compound particles half-lives

Protons, electrons, and electron neutrinos are composed exclusively of bubbles, without their corresponding anti-bubbles. This unique composition is a key factor in their exceptionally long half-lives, potentially making them stable. In contrast, particles containing both bubbles and anti-bubbles are significantly less stable. Their shorter half-lives result from the likelihood of annihilation events between bubbles and anti-bubbles within their structure, leading to their transient nature.

The neutron embeds lots of electron antineutrinos, which bind the negative charge to its inner proton. This explains the decay of the free neutron.

$$n^0 \to p^+ + e^- + \nu_e$$

12.14 Nuclear decay

In this model, nuclear decay processes emerge from the internal structure and composition of compound particles representing atomic nuclei. These instabilities trigger reorganizations via the electroweak collapse mechanism, where the sieve test result ($s2B$) determines the feasibility of transformations. The decay process is similar to the free neutron decay seen above. The perturbing particles are typically neutrinos and Z bosons that permeate the vacuum. Once again, if these perturbing particles belong to the other sector (Orbis/Umbra), then we have spontaneous decay.

This internal annihilation mechanism provides a foundational explanation for the decay processes observed in nature, aligning the model with empirical observations of nuclear instability as presented in the well-known tables of nuclides.

12.15 Heavier quarks

The framework suggests that the other quarks follow the composition of leptons, with additional neutrinos and antineutrinos, besides a variable number of propellers.

12.16 Atoms

Bound states of protons, neutrons, and electrons are naturally expected in this scenario.

12.17 The abundance of neutrinos

Despite the relatively small number of neutrinos observed in Table 4, their large cosmic abundance reflects their low mass, approximately five million times smaller than the mass of an electron. This characteristic allows neutrinos to exist in enormous quantities across the universe, even if their presence appears limited.

12.18 Partons

The inertia mechanism adopted here implies that the collective action of a large number of identical propellers produces a high-energy configuration. This stretches the propelled mass along the direction of motion, thereby forming a *parton* structure.

12.19 Quest for qubits

An electron may be conceived as a vast ensemble of dynamic *nanostates*, collectively manifesting as a single particle through their shared affinity. When two electrons share the same affinity (totally or partially), they become distinguishable only through the constraints imposed by quantization dynamics. However, if one momentarily relaxes these constraints, such a pair can be interpreted as a system of qubits—capable of exhibiting a far richer spectrum of interactions.

What, then, supports this emerging computation? Spatial interaction is facilitated by the convolution step, while the self-interference mechanism (see Sect. 7.2) introduces a form of temporal smearing that allows information to persist and influence future states.

Moreover, the atoms that host these electron structures local space via the SU(3) symmetry induced by the strong charge, further enriching the landscape of possible interactions.

This broader perspective suggests a behavior akin to an intricate network of Turing machines, each representing a potential computational pathway. The computation results are being revealed by interpreting correlations. Such an interpretation may offer new insight into the astonishing experimental results of quantum computing.

12.20 Observers and spacetime

This framework extends Wheeler's "it from bit" concept by providing a mechanism through which information constitutes physical entities, albeit from an omniscient perspective. To address the internal viewpoint, we now define inner observers or simply *observers*.

Minimal observer definition

Here, we formalize the concept of a *minimal observer*, following the framework introduced in Hatem Elshatlawy et al. [25]. A minimal observer is defined as the simplest system capable of performing observation in a functional sense, embodying core components of sensing, internal processing, and acting upon an environment within a closed feedback loop.

Definition 12.1 (Minimal Observer) *Let O be a system described by the tuple*

$$O = (X, Y, Z, f, g, B),$$

where:

- X*: Internal state space (e.g., memory or configuration states),*
- Y*: Input (sensor) space,*
- Z*: Output (action) space,*
- $f : X \times Y \to X$*: State transition function (state update rule),*
- $g : X \to Z$*: Output function (generates actions based on internal state),*
- B*: Boundary condition demarcating the internal observer from its external environment.*

The system O qualifies as a minimal observer *if it satisfies:*

- $|Y| \geq 1$*: Non-trivial sensing,*
- $|Z| \geq 1$*: Non-trivial actuation,*
- $|X| > 1$*: Non-trivial internal dynamics,*
- ***Feedback closure:*** *The output $g(x)$ must influence the environment in such a way that it alters future inputs $y \in Y$.*

This definition ensures that the observer possesses the minimal necessary structure to distinguish itself from its environment and engage in interaction through perception and response. Importantly, this framework abstracts away from higher-order capabilities such as memory, learning, or consciousness, focusing instead on the foundational structure that enables observation.

Spacetime
As established, the spacetime of this model is inherently flat and regular. However, the spacetime perceived by observers and laboratory instruments appears curved, in accordance with the principles of General Relativity. How can this apparent discrepancy be reconciled?

The resolution lies in understanding the role of propellers, which contribute both to motion and to mass-energy content. Their interactions create a framework in which observers, laboratory instruments, and the environment mutually influence one another. This interplay generates effective gravitational potentials, causing clocks to tick slower near concentrations of mass. Such behavior unveils a *timescape* scenario [26], where the perception of curved spacetime naturally emerges from relational dynamics within an underlying flat background.

Given the mutual dependence of all system components and the absence of free external variables, the model aligns with superdeterministic interpretations of reality, as discussed by Hossenfelder and Palmer [27].

13 CONCLUSION

13.1 Undecidability and long-term dynamics

Long-term dynamics involve Poincaré cycles. Because the CA can be Turing-equivalent, questions about recurrence versus chaos are undecidable in general [28, 29]. The finite state space guarantees eventual recurrence (Pigeonhole principle), but the configuration space grows exponentially with L, making detection infeasible [30, 31]. Affinity further stretches the probable recurrence time: by binding bubbles into persistent particles and aligning them into long-lived 3D islands, it reduces the effective number of independently evolving configurations and channels the dynamics through organized attractors rather than allowing rapid, unstructured mixing. Black-hole fusion may then absorb exceptional configurations ('Gardens of Eden') and drive the system toward a Poincaré cycle. Undecidability limits algorithmic predictability, so analytical tools are needed for general insights.

13.2 Falsifiability of the physical model

A direct observation that an electron physically traverses both paths in a double-slit experiment would contradict the model. Self-interference here arises from recorded lattice traces, not from simultaneous path occupation.

13.3 Novelty of the work

Compared with prior work surveyed in Sect. 2, this model introduces four main novelties:

1. A non-spatial dimension W that acts as a dynamic computational layer, enabling cross-layer convolution, a superluminal but non-signaling update mechanism, and richer interactions than Zuse's purely spatial "digital particles."

2. Six charge bits per cell, giving Lie-group-like symmetries and Noether-like conservation at a programmable level.

3. Affinity, which groups bubbles into particles and encodes self-interference histories.

4. Ontological collapse events that break strict reversibility within a Poincaré cycle, matching the effective irreversibility of quantum measurement.

The model retains a discrete lattice while capturing quantum and relativistic features; it remains exploratory and must still be validated against empirical data.

13.4 Future inclusions

Ongoing work simulates the geometric structures that emerge from these local, multi-layered updates. Preliminary tests already show that the collective dynamics of fragments produce a continuous $r \cdot \sin(r)$ density cloud, localized wave-packets, stable spiral patterns and macroscopic rotations directly from discrete bit distributions. These observations support the conjecture that the universal fabric can bridge discrete local rules and the smooth symmetries of physical spacetime.

13.5 Final thoughts

The model is an economical ontological framework[||] operating at a deep subatomic scale. Observable reality is approximate and emergent, with continuous spacetime symmetries (Lie, Lorentz, diffeomorphism, CPT, etc.) arising from the dynamics. Each particle is composed of an enormous number of bubbles—each a miniature universe—so intrinsic non-locality provides a natural substrate for superposition-like behavior and qubit analogs. The theory remains to be refined computationally and tested against experiment; whether it can become a predictive, falsifiable, *bona fide* physical theory is an open question.

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Appendix

A CALCULATION OF X AND L

Estimates for the value of X and the distance between the cells and L (see Sect. 3), the granularity of the grid, will be developed in this appendix.

The diameter of the observable universe [33] is compared to the diagonal of the lattice.

$$O = 8.8 \times 10^{26} m = \sqrt{3}\, L \cdot X. \tag{20}$$

The average photon frequency of CMB [34] is

$$[f] = 160\, GHz = 160 \times 10^{6}\, Hz,$$

which, converted to energy, gives

$$E = 1.060171224 \times 10^{-25} J.$$

The proton mass in Joules is

$$m_p = 1.503277592969 \times 10^{17} J,$$

so, the number of average photons in a proton is

$$n_p = 1.4179573 \times 10^{42}.$$

The Eddington number gives the number of protons in the universe.

$$Ed = 10^{80}.$$

Let us calculate the total contribution due to protons in terms of average photon

$$n_{pT} = 4\, Ed\, n_p$$

or

$$n_{pT} = 5.67183 \times 10^{122}.$$

On the other hand, the number of photons in the universe is

$$n_\gamma = 4 \times 10^{84},$$

which adds almost nothing to the total number due to protons

$$n_{\gamma T} = n_\gamma + n_{pT} = 5.67183 \times 10^{122}.$$

The wavelength of a single bubble in meters is

$$\lambda_0 = L \cdot X$$

and the frequency of one bubble, in Herz reads

$$f_0 = \frac{c}{L \cdot X},$$

helping to find the number of bubbles in the average photon

$$n_b = \frac{[f]}{f_0} = \frac{160 \times 10^{6}\, L \cdot X}{c}.$$

So, the amount of bubbles due to all photons is

$$n_{bT} = n_b\, n_{\gamma T} = 2\frac{5.67183 \times 10^{122} \times 160 \times 10^6\, L \cdot X}{c}$$

and

$$n_{bT} = L^3.$$

We then have

$$\frac{1814.9856 \times 10^{128}\, X}{c} = L^2. \tag{21}$$

Using Eqn. (20) and Eqn. (21), we obtain

$$X \approx 7.52831 \times 10^{-24} m$$

and

$$L \approx 6.74877 \times 10^{49}.$$

Assuming that the Planck length is a valid length scale, let us recalculate Ed for $X = l_p$.

$$Ed = \frac{\sqrt{3}\, c\, O^2}{1920\, n_p\, l_p^3} = 2.18816 \times 10^{113}.$$

Then we finally obtain

$$X \approx 1.616199 \times 10^{-35} m = l_p$$

and

$$L \approx 6.95410 \times 10^{60}.$$